

Jiahui Qu

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EDUCATION

London School of Economics and Political Science

MSc Statistics (Financial Statistics)

Selected marks: Artificial Intelligence (87/100); Managing and Visualising Data (84/100); Reinforcement Learning (83/100)

London, UK

Sep. 2025 – Mar. 2027

University of Edinburgh

BSc (Hons) Mathematics and Statistics

Exchange: City University of Hong Kong, *Computing Mathematics* (2023–2024)

GRE: 338/340 (V168, Q170), Analytical Writing 4.0.

Edinburgh, UK

Sep. 2021 – Jun. 2025

PUBLICATIONS & MANUSCRIPTS

- [1] **Jiahui Qu**, Yifang Qin, Boyang Zheng, Ziyi Zhou. “Calibrate Once, Choose the Beam: A Predictive Regime Test for Same-LM Search Guidance and Pruning.” *ICML 2026 LM4Plan Workshop (accepted)*.
- [2] **Jiahui Qu**, Yifang Qin, Boyang Zheng, Ziyi Zhou. “Calibrate Once, Choose the Beam: A Pre-Deployment Compute-Allocation Rule for Same-LM Multimodal Search.” *ICML 2026 SCALE Workshop (accepted)*.
- [3] **Jiahui Qu**, Boyang Zheng, Yifang Qin, Yudi Cao, Jiayu Fang. “Screen, Confirm, Then Search: Risk-Controlled Pre-Deployment Qualification for LM-Guided Search.” *TMLR (under review)*.
- [4] **Jiahui Qu**, Shizhao Xiong, Jiayu Fang, et al. “When Better Scores Select Worse Boards: Replacement-Level Audits for Teamfight Tactics Search.” *Under review*.
- [5] **Jiahui Qu**, Shizhao Xiong, Yifang Qin, et al. “When Evidence Drifts: Evidence-State Routing and Guarded Revision Release for Long-Form Factual Generation.” *Under review*.

RESEARCH EXPERIENCE

Lasso-to-Learn: Adaptive AI Intervention for Learning

Dec. 2025 – Present

Independent Research Project | YNotes

- Model learner state and estimate the counterfactual effects of **wait**, **probe**, and **hint** actions during active reading and note-taking, with the goal of learning minimally sufficient intervention policies.
- Design source-anchored interactions that connect learner-selected material with contextual feedback and follow-up practice; design counterfactual evaluations and human-subject studies comparing alternative intervention policies.

A Study on Hoeffding’s D-Test for Independence

May. 2024 – Aug. 2024

Funded Research Student | University of Edinburgh | Supervised: William Scott

- Studied Hoeffding’s D-statistic for detecting non-monotonic dependence and investigated its theoretical relationship with Even-Zohar’s representation.
- Applied dependence testing and bootstrap inference to financial-stability data from 13 African countries, with statistical and machine-learning baselines for systemic-crisis prediction.

Moments of Random Matrices and Combinatorics

Apr. 2023 – Sep. 2023

Funded Research Student | University of Edinburgh | Supervised: Theo Assiotis

- Studied asymptotic questions related to the Keating–Wong conjecture for moments of random matrices and log-correlated fields.
- Derived and computationally verified expressions involving Gamma-function sums, using symbolic and numerical experiments to examine large-N asymptotics.

SOFTWARE & SYSTEMS

YNotes — Founder & Developer

2025 – Present

AI-native iPad notebook for source-grounded learning | App Store | ynotesapp.com

- Designed and shipped a handwriting-first learning workflow in which learners can lasso source material, request contextual tutoring or solutions, keep AI explanations alongside the original content, and generate follow-up practice without leaving the learning workspace.
- Built and deployed the end-to-end system; released YNotes on the **App Store** and use it as the experimental platform for Lasso-to-Learn.

HONOURS & AWARDS

- LSE Annual Scholarship**, London School of Economics and Political Science 2025
- Undergraduate Vacation Scholarship**, University of Edinburgh 2023 - 2024
- Kaggle Silver Medal — GoDaddy Microbusiness Density Forecasting**, Top 3% 2023
- Edinburgh Leadership Award**, University of Edinburgh 2023

LEADERSHIP & MENTORING

Peer Support Leader, University of Edinburgh 2022 - 2023

- Mentored junior students through peer-learning sessions and academic activities, supporting their transition to university-level mathematics and independent study.

Department Lead, Lambda — AI & Computing Society, University of Edinburgh 2021 – 2022

- Organised an AI/computing hackathon for 50+ participants, coordinating student teams, and technical activities.

PROFESSIONAL EXPERIENCE

Guoyuan Securities Jun. 2024 – Sep. 2024

Quantitative Equity Investment Intern | Wuhan, China

- Developed and backtested ETF-rotation strategies using momentum, trend, and cross-asset correlation signals; evaluated portfolio returns, Sharpe ratio, and drawdown.

Guotai Junan Securities Jun. 2022 – Sep. 2022

Quantitative Researcher Intern | Shanghai, China

- Built portfolio and factor analytics in Python and R using market and valuation data; conducted return/risk attribution and quantitative equity research.

PATENT

Data Classification Framework for Financial Economics 2023

Patent No. **ZL 2023 3 0301344.6**

TECHNICAL SKILLS

Programming: Python, R, MATLAB, Swift/SwiftUI, SQL.

Machine Learning & Statistics: PyTorch, Hugging Face, scikit-learn, LightGBM, statistical modelling, bootstrap inference, experimental design.

Systems & Tools: PencilKit, Git, LaTeX, Cloudflare Workers.

Languages: Mandarin Chinese (native), English (fluent).